

MAT 3150 — Homework 3

Problem 4 Walkthrough: Averages of an Increasing Sequence

Goal. Let $(x_n)_n$ be an increasing sequence of positive real numbers and define

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Prove that (y_n) is increasing as well.

What “increasing” means

A sequence (a_n) is increasing if $a_n \leq a_{n+1}$ for every n . (If your course uses *strictly* increasing, replace $\leq$ by $<$ everywhere below; the argument is the same.)

The standard way to prove it: show that

$$a_{n+1} - a_n \geq 0 \quad \text{for every } n.$$

With fractions, that means: put $a_{n+1} - a_n$ over a common denominator, note the denominator is positive, and show the numerator is ≥ 0 .

Notation that makes this problem manageable. Let

$$S_n = x_1 + x_2 + \cdots + x_n, \quad \text{so} \quad y_n = \frac{S_n}{n}.$$

The one fact about S_n you’ll need:

$$S_{n+1} = S_n + x_{n+1}.$$

Worked example: $a_n = \frac{n}{n+1}$ is increasing

Compute the difference. Common denominator $(n+1)(n+2)$:

$$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{n^2 + 2n + 1 - n^2 - 2n}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}.$$

Sign check. The numerator $1 > 0$ and the denominator $(n+1)(n+2) > 0$, so $a_{n+1} - a_n > 0$.

Proof. For every $n \in \mathbb{N}$,

$$a_{n+1} - a_n = \frac{1}{(n+1)(n+2)} > 0,$$

since the numerator and denominator are both positive. Hence $a_n < a_{n+1}$ for every n , so (a_n) is increasing. $\square$

Pattern: difference $\rightarrow$ common denominator $\rightarrow$ simplify $\rightarrow$ show numerator and denominator have the right signs.

Your turn: $y_n = S_n/n$

1. Set up the difference. ✓ (done)

$$y_{n+1} - y_n = \frac{S_{n+1}}{n+1} - \frac{S_n}{n}.$$

2. Common denominator. ✓ (done) Multiply the first fraction by $\frac{n}{n}$ and the second by $\frac{n+1}{n+1}$:

$$y_{n+1} - y_n = \frac{n S_{n+1} - (n+1) S_n}{n(n+1)}.$$

Denominator sign. ✓ (done) $n(n+1) > 0$ since $n \geq 1$. So the sign of $y_{n+1} - y_n$ is the sign of the numerator.

Expand the numerator. ✓ (done) Substitute $S_{n+1} = S_n + x_{n+1}$:

$$n S_{n+1} - (n+1) S_n = (n S_n + n x_{n+1}) - (n S_n + S_n).$$

3. Cancel. ✓ (done) Distribute the minus sign over both terms in the second bracket:

$$(n S_n + n x_{n+1}) - (n S_n + S_n) = n S_n + n x_{n+1} - n S_n - S_n.$$

The first term $+n S_n$ and the third term $-n S_n$ cancel ✓. The two terms that remain:

$$= n x_{n+1} - S_n.$$

So

$$y_{n+1} - y_n = \frac{n x_{n+1} - S_n}{n(n+1)}.$$

Step 4 — where “increasing” is used.

Numerical example first. Take the increasing sequence $x_1 = 2$, $x_2 = 5$, $x_3 = 7$, $x_4 = 10$ and $n = 3$, so $x_{n+1} = x_4 = 10$.

- Each earlier term is at most the new one: $2 \leq 10$, $5 \leq 10$, $7 \leq 10$.
- Add them: $\underbrace{2 + 5 + 7}_{S_3=14} \leq \underbrace{10 + 10 + 10}_{3 \cdot 10=30}$.
- Numerator: $3 \cdot 10 - 14 = 16 \geq 0$.
- Check: $y_3 = 14/3 \approx 4.67$, $y_4 = 24/4 = 6$, and $\frac{16}{3 \cdot 4} = \frac{4}{3} = y_4 - y_3$. ✓

Example → general.

	Example ($n = 3$)	General
New term	$x_4 = 10$	x_{n+1}
Earlier terms	$2, 5, 7$	$x_1, \dots, x_n$
Each earlier $\leq$ new	$2 \leq 10, 5 \leq 10, 7 \leq 10$	$x_k \leq x_{n+1}, k = 1, \dots, n$
Sum of left sides	$2 + 5 + 7 = S_3$	$x_1 + \dots + x_n = S_n$
Sum of right sides	$10 + 10 + 10 = 3 \cdot 10$	$x_{n+1} + \dots + x_{n+1} = n x_{n+1}$
Result	$S_3 \leq 3 \cdot 10$	$S_n \leq n x_{n+1}$

4. Show the numerator is ≥ 0 . ✓ (done) Since (x_n) is increasing,

$$x_k \leq x_{n+1} \quad \text{for each } k = 1, \dots, n.$$

Adding these n inequalities: the left sides sum to S_n , and the right side is n copies of x_{n+1} , so

$$S_n \leq n x_{n+1}.$$

Subtracting S_n from both sides,

$$n x_{n+1} - S_n \geq 0.$$

5. Conclude. ✓ (done) Numerator ≥ 0 and denominator $n(n+1) > 0$, so

$$y_{n+1} - y_n = \frac{n x_{n+1} - S_n}{n(n+1)} \geq 0,$$

i.e. $y_n \leq y_{n+1}$ for every n .

In your own words: why is $S_n \leq n x_{n+1}$? Finish the sentence: “Each of the n terms $x_1, \dots, x_n$ is at most _____, so their sum is at most _____.”

6. The proof. Fill in the template:

Let $S_n = x_1 + \dots + x_n$, so $y_n = S_n/n$ and $S_{n+1} = S_n + x_{n+1}$. For every n ,

$$y_{n+1} - y_n = \frac{\quad}{n(n+1)}.$$

Since (x_n) is increasing, $x_k \leq x_{n+1}$ for $k = 1, \dots, n$; adding these gives $S_n \leq \underline{\hspace{2cm}}$. Hence the numerator is ≥ 0 . The denominator $n(n+1)$ is positive. Therefore $y_{n+1} - y_n \geq 0$, i.e. $y_n \leq y_{n+1}$ for every n , so (y_n) is increasing.

Check yourself.

- Where did you use that (x_n) is increasing?
- Did you use that the x_n are *positive*? (It's fine if you didn't — just notice.)